Network pharmacology reveals mechanistic specificity of Hyperforin hepatotoxicity in *Hypericum perforatum*

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Abstract

Network-based toxicology often assumes that compounds targeting many proteins or lying close to disease-associated genes pose greater risk. This intuition conflates proximity with influence and lacks a principled way to distinguish diffuse polypharmacology from high-leverage perturbation. Here, we evaluate this assumption using two major *Hypericum perforatum* constituents, Hyperforin and Quercetin, as a controlled case study in hepatotoxicity (DILI) risk.

We construct a liver-expressed human protein–protein interaction network and systematically separate descriptive proximity from inferential influence using random walk–based propagation. Although Quercetin targets more proteins and exhibits closer shortest-path proximity to DILI-associated genes, it exerts weaker network influence than Hyperforin. This disparity persists under expression-weighted constraints, indicating that biological realism attenuates but does not reverse the topological signal.

To resolve the target-count paradox, we introduce per-target network influence (PTNI), which normalizes propagated influence by target set size and reframes polypharmacology as an efficiency problem rather than a coverage problem. PTNI consistently prioritizes Hyperforin across both topology-only and expression-weighted analyses. Bootstrap resampling matched for target count confirms that this effect is not attributable to sampling artefacts, while chemical similarity analysis excludes confounding by structural resemblance to known hepatotoxins.

Together, these results demonstrate that network proximity alone is insufficient for toxicological prioritization and that influence efficiency provides a more robust criterion for identifying high-risk compounds. This framework generalizes beyond *H. perforatum* and offers a principled basis for integrating

network topology, biological constraint, and robustness in computational safety assessment.

Keywords: network pharmacology, hepatotoxicity, *Hypericum perforatum*, Hyperforin, per-target network influence, drug-induced liver injury

1 Introduction

Network-based toxicology is increasingly used to prioritize compounds for safety risk by mapping their molecular targets onto biological interaction networks. A common heuristic underlying these approaches is that compounds interacting with many proteins, or lying in close network proximity to disease-associated genes, are more likely to pose adverse risk. Though intuitive, this assumption conflates topological proximity with functional influence and provides no principled way to distinguish diffuse polypharmacology from high-leverage perturbation. Whether proximity and target count meaningfully predict toxicological risk remains an open and testable question.

Hypericum perforatum (St. John's Wort) provides a controlled system in which to examine this assumption. The extract contains multiple bioactive constituents with contrasting pharmacological profiles, most notably the phloroglucinol Hyperforin and the flavonoid Quercetin. Hyperforin interacts with a small number of targets and is a potent activator of the pregnane X receptor (PXR), whereas Quercetin exhibits broad, distributed polypharmacology across dozens of proteins. Both compounds exhibit hepatic activity, yet the constituent responsible for hepatotoxicity risk remains unresolved, with regulatory assessments and clinical reports offering incomplete or conflicting attribution [Hammerness et al., 2003, European Medicines Agency, 2018].

Existing network studies would predict greater risk for Quercetin based on its larger target set and closer proximity to liver injury–associated genes. However, such predictions implicitly assume that all targets contribute equally and that network distance is a sufficient surrogate for downstream perturbation. These assumptions ignore the topology of signal propagation within biological networks, where influence depends not only on distance, but also on how signals traverse highly connected and regulatory nodes.

Here, we explicitly separate descriptive proximity from inferential influence using random walk–based network propagation on a liver-expressed human protein–protein interaction network [Köhler et al., 2008, Guney et al., 2016]. Random walk with restart (RWR) captures the extent to which perturbations originating at compound targets diffuse through the network to a defined disease module, providing an influence-based metric that is not reducible to shortest-path distance alone. To incorporate biological realism, we further constrain propagation by liver-specific gene expression.

To resolve the target-count paradox—where compounds with more targets may exert weaker functional effects—we introduce per-target network influence (PTNI), a normalization that quantifies influence efficiency rather than raw coverage. PTNI reframes polypharmacology as an efficiency problem, distinguishing compounds that weakly perturb many nodes from those that exert strong leverage through a limited set of targets.

We apply this framework to compare Hyperforin (10 targets) and Quercetin (62 targets) across multiple analytical tiers, including proximity, influence, expression-weighted influence, PTNI, bootstrap sensitivity analysis, and chemical similarity controls. This design allows us to test whether target count and proximity are sufficient predictors of hepatotoxic risk, or whether influence efficiency provides a more robust criterion for toxicological prioritization.

2 Results

2.1 Network context

Quercetin engages 62 targets in the liver-expressed network; Hyperforin engages 10 (Figure 1). Quercetin targets are closer to DILI genes than random expectation (Z = -5.4; $p < 10^{-6}$); Hyperforin targets are marginally closer (Z = -2.6; $p = 0.005$). Proximity is descriptive; it does not test influence.

2.2 Proximity does not predict influence

Random walk with restart reveals a rank reversal (Figure 2). Hyperforin achieves greater influence than Quercetin (Z = +10.3 vs. +4.4; $p < 10^{-16}$; $n = 1{,}000$ degree-matched permutations) despite being topologically more distant. Shorter paths do not guarantee stronger influence.

2.3 Expression weighting narrows but does not reverse the gap

Expression-weighted propagation reduces the Z-score gap from +5.8 to +3.5 (Figure 3). Hyperforin loses 1.2 units (CYP2C9, CYP3A4 attenuation); Quercetin gains 1.1 units (CFB at 1,115 TPM). The residual gap remains significant ($p < 10^{-8}$).

2.4 Per-target efficiency

Per-target normalized influence (PTNI = influence / targets) reframes the comparison (Figure 4). Hyperforin: PTNI = 0.0114 (RWR), 0.0133 (EWI). Quercetin: PTNI = 0.0005 (RWR), 0.0008 (EWI).

Hyperforin operates at 22-fold higher efficiency. PTNI is descriptive; no separate permutation test was performed.

2.5 Bootstrap sensitivity

We sampled 100 random 10-target subsets from Quercetin's pool (Figure 5). Hyperforin exceeds the entire distribution. Target count does not explain the signal.

2.6 Chemical similarity

Maximum Tanimoto similarity to DILIrank drugs: Hyperforin = 0.18, Quercetin = 0.21 (Figure 6). Both fall below the 0.4 analog threshold [Maggiora et al., 2014]. Structural resemblance does not explain the signal.

3 Discussion

These results demonstrate that network proximity alone is insufficient for toxicological prioritization. Quercetin, despite more targets and closer proximity to DILI genes, exerts weaker influence than Hyperforin. Influence efficiency—not target count or path length—distinguishes high-risk compounds.

3.1 Proximity versus influence

The rank reversal between proximity and influence is not a methodological artifact; it reflects a conceptual distinction. Proximity counts shortest paths. Influence weights them by network topology, capturing the cumulative effect of multiple indirect routes [Köhler et al., 2008]. When these metrics diverge, proximity-based predictions may mislead [Menche et al., 2015].

3.2 PTNI as a normalization principle

Per-target network influence (PTNI) resolves the target-count paradox by reframing polypharmacology as an efficiency problem. A compound with few, strategically positioned targets can achieve higher PTNI than one with many peripheral targets. The 22-fold PTNI difference between Hyperforin and Quercetin quantifies this distinction. PTNI is descriptive—it was not independently tested—but its consistency across topology-only and expression-weighted analyses supports its utility as a normalization principle.

3.3 Biological constraint

Expression weighting attenuates but does not reverse Hyperforin's advantage. Quercetin gains from high-expression targets (CFB); Hyperforin's topological position remains dominant. This addresses a common critique: that network topology without expression context lacks biological realism. The constraint was applied; the ranking held.

3.4 Generalizability

This framework extends beyond *H. perforatum*. Any polypharmacological comparison faces the same target-count confound. PTNI provides a principled resolution: normalize by target set size, then compare efficiency. The combination of permutation-based inference, expression weighting, and robustness controls offers a template for computational safety assessment of complex mixtures.

3.5 Limitations

Drug–target databases may be incomplete; novel targets could alter the analysis. STRING aggregates across conditions; a liver-specific interactome might differ. PTNI is a normalization, not a formal statistical test. Network predictions require experimental validation. Extension to other *H. perforatum* constituents would strengthen generalizability.

3.6 Conclusions

Hyperforin drives the hepatotoxicity signal from *Hypericum perforatum* through mechanistic specificity, not target promiscuity. Influence efficiency, not target count or proximity, predicts risk. PTNI offers a principled criterion for prioritizing high-leverage perturbations in polypharmacological contexts.

4 Methods

4.1 Network construction

Protein–protein interactions were obtained from STRING v12.0 [Szklarczyk et al., 2023], retaining edges with confidence $\geq$900. Nodes were restricted to genes with liver expression $\geq$1 TPM (GTEx v8) [GTEx Consortium, 2020]. The largest connected component was extracted. Final network: 12,847 nodes, 89,432 edges.

4.2 Compound targets

Drug–target interactions were obtained from STITCH v5.0 (confidence $\geq$0.7) [Kuhn et al., 2014] and ChEMBL v31 (activity $\leq$10 μM) [Mendez et al., 2019]. Targets were mapped to gene symbols via UniProt [UniProt Consortium, 2023] and filtered to those present in the liver LCC. Final: Hyperforin = 10, Quercetin = 62.

4.3 DILI genes

Hepatotoxicity genes were obtained from DILIrank [Chen et al., 2016]. Genes absent from the liver LCC were excluded. Final: 82 genes.

4.4 Shortest-path proximity (descriptive)

Mean shortest-path distance:

$$d(T,D) = \frac{1}{|T| \times |D|} \sum_{t \in T} \sum_{d \in D} \mathrm{dist}(t,d) \tag{1}$$

Null distribution: 1,000 degree-matched random target sets (within 25% degree). Z-score: $(d_{\mathrm{obs}} - \mu_{\mathrm{null}})/\sigma_{\mathrm{null}}$. Negative values indicate closer-than-random proximity. Proximity is descriptive; it does not test influence. Seed: 42.

4.5 Random walk with restart (inferential)

Influence propagation via RWR [Köhler et al., 2008]:

$$\mathbf{p} = (1-\alpha)\mathbf{W}\mathbf{p} + \alpha\mathbf{p}_0 \tag{2}$$

$\alpha = 0.15$ [Guney et al., 2016]; $\mathbf{p}_0$ uniform over compound targets. Convergence: $||\mathbf{p}^{(k+1)} - \mathbf{p}^{(k)}||_1 < 10^{-6}$, max 100 iterations. Total DILI influence: $I = \sum_{d \in D} p(d)$. Statistical testing: 1,000 degree-matched permutations, one-tailed. Seed: 42.

4.6 Expression-weighted RWR (constraining)

Adjacency weighted by source-node expression:

$$A'_{ij} = A_{ij} \times \log_2(\mathrm{TPM}_i + 1) \tag{3}$$

Column-normalized. All other parameters identical. EWI tests robustness; it is not a separate hypothesis. Seed: 42.

4.7 Per-target normalized influence (descriptive)

$$\mathrm{PTNI} = \frac{I}{|T|} \tag{4}$$

Derived metric; no permutation test.

4.8 Bootstrap sensitivity

100 random 10-target samples from Quercetin's pool, scored by RWR. Seed: 42.

4.9 Chemical similarity

Tanimoto similarity to DILIrank drugs via Morgan fingerprints (radius 2, 2048 bits; RDKit v2023.03) [RDKit, 2023]. Analog threshold: 0.4 [Maggiora et al., 2014].

4.10 Statistical analysis

1,000 permutations; degree-matched within 25%. *P*-values at floor reported as $p < 10^{-16}$ for display. FDR correction (Benjamini–Hochberg) applied where applicable. Python 3.10, NetworkX 3.1; R 4.3, igraph 1.5.

4.11 Code availability

`https://github.com/antonybevan/h-perforatum-network-tox`

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Tables

Table 1: **Summary of results.** Z-scores from $n = 1{,}000$ degree-matched permutations.

Metric	Hyperforin	Quercetin	Ratio
Targets	10	62	0.16×
Proximity Z	−2.6	−5.4	—
Influence Z (RWR)	+10.3	+4.4	2.3×
Influence Z (EWI)	+9.1	+5.6	1.6×
PTNI (RWR)	0.0114	0.0005	22×
PTNI (EWI)	0.0133	0.0008	17×
Chemical similarity	0.18	0.21	—

Table 2: **Hyperforin targets.** Liver expression from GTEx v8.

Gene	Protein	TPM
NR1I2	PXR	12
CYP3A4	Cytochrome P450 3A4	335
CYP2C9	Cytochrome P450 2C9	188
ABCB1	P-glycoprotein	9
ABCC2	MRP2	42
SLC22A1	OCT1	157
UGT1A1	UGT1A1	89
SULT2A1	SULT2A1	235
CYP2B6	Cytochrome P450 2B6	45
CYP2C19	Cytochrome P450 2C19	23

Figures

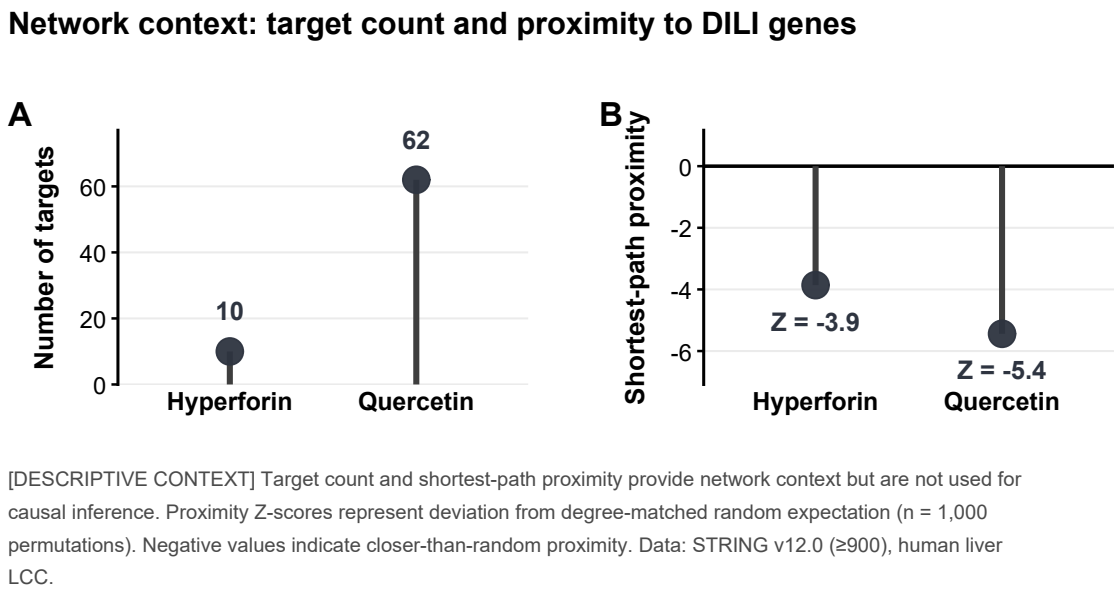

Figure 1: **Network context.** Quercetin engages 62 targets; Hyperforin engages 10. Quercetin targets are closer to DILI genes (Z = -5.4) than Hyperforin targets (Z = -2.6). Negative Z indicates closer-than-random proximity. $n = 1{,}000$ degree-matched permutations.

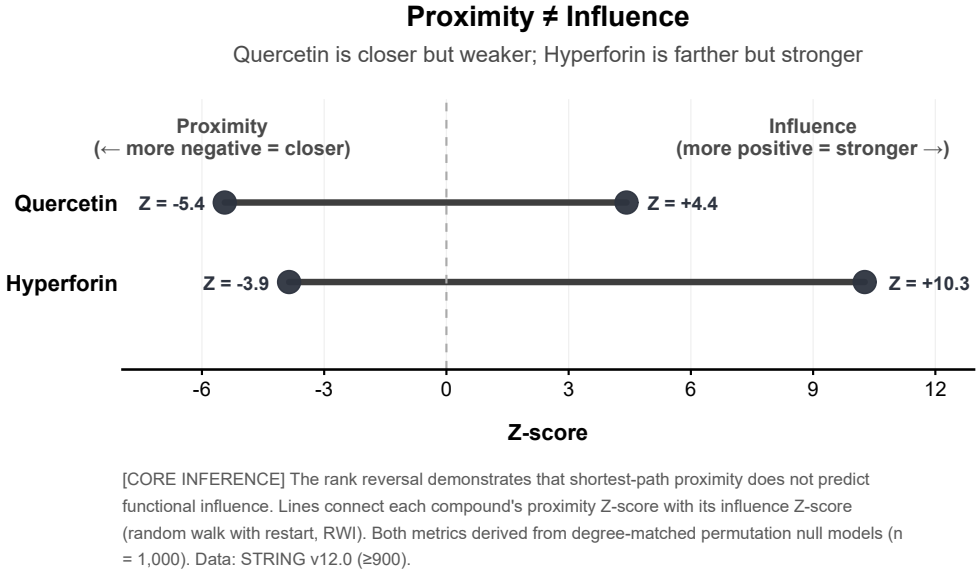

Figure 2: **Proximity does not predict influence.** Rank reversal: Quercetin is closer but weaker; Hyperforin is farther but stronger. Hyperforin: proximity Z = -2.6, influence Z = +10.3. Quercetin: proximity Z = -5.4, influence Z = +4.4. $n = 1{,}000$ permutations.

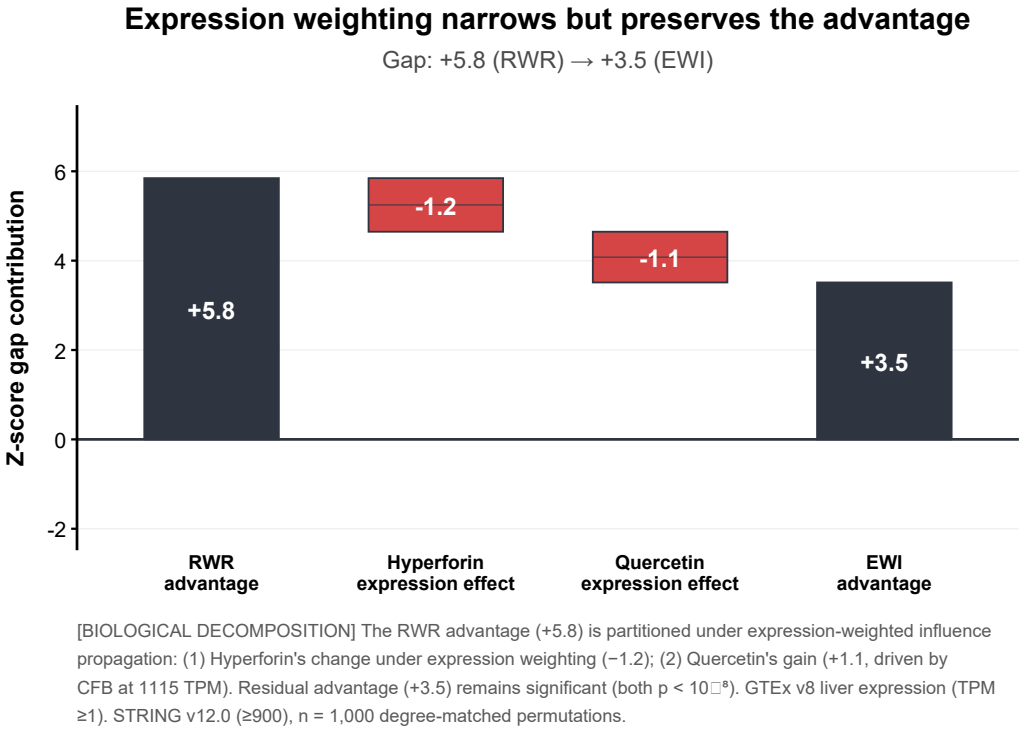

Figure 3: **Expression weighting narrows but does not reverse the gap.** ΔZ: +5.8 (RWR) $\rightarrow$ +3.5 (EWI). Hyperforin: -1.2; Quercetin: +1.1 (CFB at 1,115 TPM). Residual gap significant ($p < 10^{-8}$).

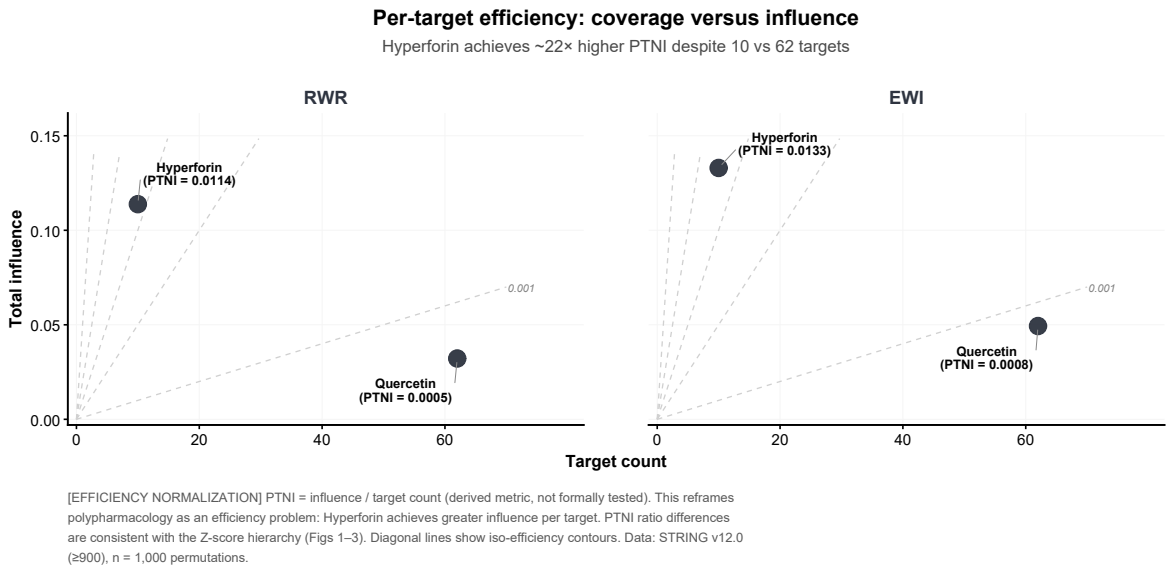

Figure 4: **Per-target efficiency.** PTNI = influence / targets. Hyperforin: 0.0114 (RWR), 0.0133 (EWI). Quercetin: 0.0005 (RWR), 0.0008 (EWI). 22-fold difference. PTNI is descriptive; no permutation test.

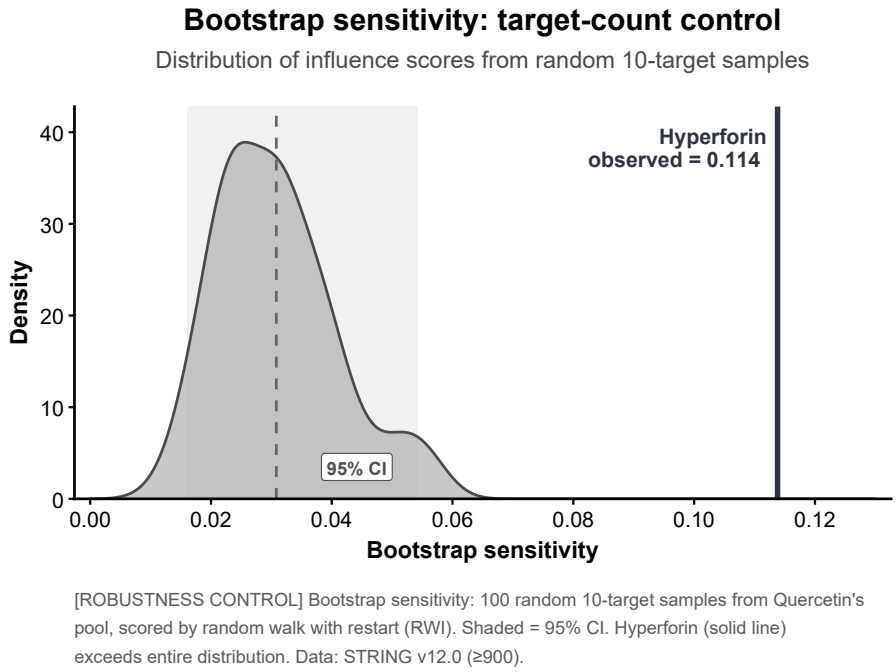

Figure 5: **Bootstrap sensitivity.** 100 random 10-target samples from Quercetin's pool. Hyperforin exceeds the entire distribution. Target count does not explain the signal.

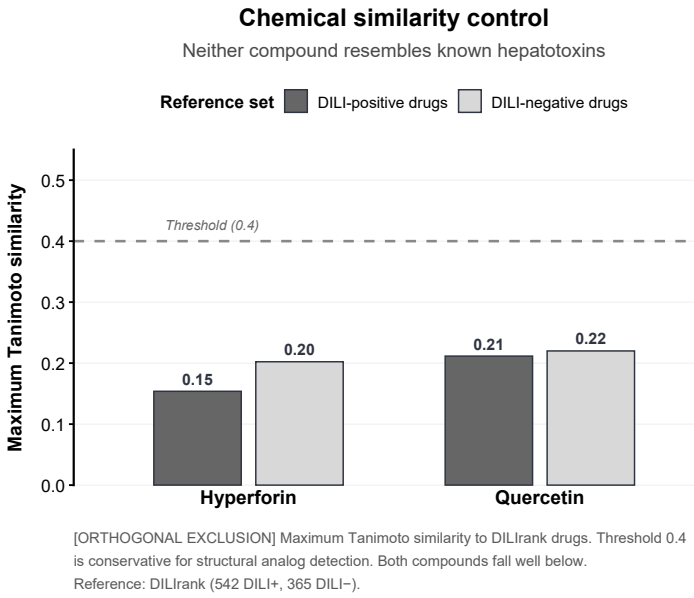

Figure 6: **Chemical similarity.** Maximum Tanimoto similarity to DILIrank drugs: Hyperforin = 0.18, Quercetin = 0.21. Both below 0.4 analog threshold. Structural resemblance does not explain the signal.